DISEASE ANALYSIS USING HOSPITAL DATA

A Project Report submitted in partial fulfillment of the

requirements for the award of degree of

BATCHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted

By

Y. Venkat (24B11AI460)

Y . Chakri (24B11AI462)

Y . chaitanya (24B11AI470)

B . Koushik (25B21AI046)

Under the esteemed guidance of

Ms. Pravallika , Qualification.

Designation.

ADITYA UNIVERSITY

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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(Formerly Aditya Engineering College (A))

ADITYA UNIVERSITY

(Formerly Aditya Engineering College(A))

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

ADITYA
UNIVERSITY

CERTIFICATE

This is to certify that the project work entitled " DISEASE ANALYSIS USING HOSPITAL DATA" is being submitted by

Y. Venkat (24B11AI460)

Y . Chakri (24B11AI462)

Y . chaitanya (24B11AI470)

B . Koushik (25B21AI046)

in partial fulfillment of the requirements for award of the B.Tech. degree in Artificial Intelligence and Machine Learning.

Project Guide

Ms. Pravallika, Qualification.
Designation.

Head of the Department

Dr. M. Venkatesh, M.Tech., Ph.D.
Assoc. Prof., AIML Dept.

DECLARATION

We hereby declare that the project entitled **"Customer Churn prediction and Sentiment Analysis"** is a genuine project. This work has been submitted to the **ADITYA UNIVERSITY (Formerly Aditya Engineering College (A)),** Surampalem in partial fulfillment of the **B.Tech.** degree**.** We further declare that this project work has not been submitted in full or part of the award of any degree or any other educational institutions.

by

Y. Venkat (24B11AI460)

Y. Chakri (24B1AI42)

Y. Chaitanya (24B11AI470)

B. Koushik (25B21AI046)

ACKNOWLEDGEMENT

It is with immense pleasure that we would like to express us indebted gratitude to our project Guide **Ms. Pravallika , Designation.,** who has guided us a lot and encourage in every step of the project work, his valuable moral support and guidance throughout the project helped us a greater extent.

We owe our sincere gratitude to our project coordinator **Mr. Project Coordinator Name, Designation.,** for providing a great support and guidance throughout the project.

We are grateful to **Dr. Maganti Venkatesh, Assoc. Prof.** and **HoD,** for inspiring us all the way and for arranging all the facilities and resources needed for our project.

We wish to thank **Dr. M.V. Rajesh, Assoc. Dean SoE.** for the encouragement and support during the course of my project.

We would like to extend our sincere thanks to **Dr. A. Ramesh, Pro Vice-Chancellor (Engg. & Sci.), Dr. G. Suresh, Registrar, Dr. S.Rama Sree, Pro Vice-Chancellor (Academics), Dr. M.B. Srinivas, Vice-Chancellor, Dr. M. Sreenivasa Reddy, Deputy Pro-Chancellor and Management, Aditya University** for unconditional support for providing us the best infrastructural facilities and state of the art laboratories during our project.

Not to forget, Parents **Faculty, Teaching, Non-Teaching Staff and our Friends** who have directly or indirectly supported us in completing this project on time.

ABSTRACT

The rapid growth of healthcare systems has resulted in the generation of large volumes of hospital data, including patient records, disease details, age, gender, and medical information. Analyzing this data manually is time-consuming and inefficient, making it difficult for healthcare institutions to understand disease patterns and patient health trends accurately. This may affect effective decision-making and healthcare planning.

This project focuses on analyzing hospital data using data mining techniques to identify frequently occurring diseases and understand their distribution among different patient groups. The main objective is to analyze disease patterns based on patient demographics such as age and gender and to extract meaningful insights from the dataset.

Before analysis, data preprocessing techniques such as data cleaning, handling missing values, removal of duplicate records, and standardization are applied to improve data quality and consistency. The implementation is carried out using tools like WEKA and Python libraries for effective data analysis and visualization.

For disease analysis, classification algorithms such as Decision Tree and Naïve Bayes are applied to identify disease patterns, while K-Means clustering is used to group patients with similar health conditions. These techniques help in better understanding of disease distribution and patient characteristics.

The results of this project help in identifying the most common diseases and analyzing their variation across different age groups and genders. Model performance and analysis accuracy are evaluated to ensure reliable results. The findings demonstrate that data mining techniques provide valuable insights into healthcare data.

These insights can assist hospitals in improving healthcare services, planning resources efficiently, and taking preventive measures. Overall, this project shows how data mining techniques can transform raw hospital data into meaningful information for better healthcare decision-making.

Keywords

Disease Analysis, Hospital Data, Data Mining, Data Preprocessing, Patient Records, Disease Patterns, Classification, Clustering, Decision Tree, Naïve Bayes, K-Means Clustering, WEKA, Healthcare Data Analytics, Visualization, Accuracy

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1: INTRODUCTION

1.1 Introduction:

In today's digital era, healthcare systems generate a massive amount of data through patient records, disease reports, medical tests, and hospital management systems. Analyzing this large volume of data has become essential for understanding disease patterns and improving healthcare services. One of the major challenges faced by hospitals is identifying frequently occurring diseases and understanding their distribution among different groups of patients.

Diseases can vary based on factors such as age, gender, lifestyle, and environmental conditions. However, identifying these patterns manually is difficult due to the large volume and complexity of hospital data. This is where data mining techniques play an important role in extracting useful information and identifying meaningful patterns from the data.

In addition to identifying common diseases, analyzing patient data helps in understanding how diseases spread among different age groups and genders. This enables healthcare providers to take preventive measures, plan resources efficiently, and improve patient care.

By applying data mining techniques such as classification and clustering, hospitals can analyze disease trends effectively. These techniques help in grouping similar patient records and identifying patterns that support better healthcare decision-making.

1.2 Problem Statement

In today's digital era, healthcare systems generate a massive amount of data through patient records, disease reports, laboratory results, and medical history. Analyzing this large volume of hospital data is essential for understanding disease patterns and improving healthcare services. However, extracting meaningful insights from such data is a major challenge for healthcare institutions.

Identifying frequently occurring diseases and analyzing their distribution among different patient groups is difficult when done manually due to the size and complexity

of the data. Diseases may vary based on factors such as age, gender, and other health conditions, making analysis more complicated.

Traditional methods of data analysis are often time-consuming and less effective in handling large-scale hospital datasets. Therefore, there is a need for advanced data mining techniques to analyze patient data, identify disease patterns, and generate useful insights.

By applying data mining techniques such as classification and clustering, it becomes possible to analyze disease trends, group similar patient records, and support better healthcare decision-making. This helps hospitals in improving treatment planning, resource allocation, and preventive healthcare measures.

1.3 Objectives of the Project

- Collect hospital patient datasets containing details such as Patient ID, Age, Gender, and Disease
- Perform data preprocessing (handling missing values, removing duplicates, cleaning and standardizing data)
- Convert categorical data (such as Gender and Disease) into suitable formats for analysis
- Apply feature selection and normalization techniques to improve analysis accuracy
- Apply Decision Tree algorithm to classify and analyze disease patterns
- Apply Naïve Bayes algorithm for disease classification based on patient data
- Apply K-Means clustering to group patients based on similar health conditions
- Perform disease frequency analysis to identify the most common diseases
- Evaluate analysis performance using accuracy and other evaluation metrics
- Visualize results using bar charts and pie charts for better understanding
- Analyze key factors influencing disease occurrence

- Provide insights to support healthcare planning and decision-making

1.4 Scope of the Project

- Can be used by hospitals and healthcare institutions to analyze patient data and disease patterns
- Helps in identifying frequently occurring diseases
- Assists in understanding disease distribution across different age groups and genders
- Enables visualization of disease patterns using charts and graphs
- Helps in improving healthcare planning and resource management
- Can be applied to various healthcare datasets and hospital systems
- Capable of handling large volumes of structured hospital data
- Supports data-driven decision-making in healthcare
- Can be extended with advanced data mining and machine learning techniques
- Useful for improving patient care and preventive healthcare measures

2: SYSTEM OVERVIEW

2.1 Existing System

In the current healthcare environment, hospitals collect large amounts of patient data through medical records, laboratory reports, and hospital management systems. This data includes patient details such as age, gender, disease type, and medical history.

In most cases, this data is analyzed manually or using basic analytical tools. Manual analysis requires significant time and effort, especially when dealing with large datasets. Due to this, it becomes difficult for healthcare institutions to identify disease patterns and trends effectively.

The existing system lacks efficient methods to analyze disease distribution automatically. It also fails to provide clear insights into frequently occurring diseases and their relationship with patient demographics, making it difficult for hospitals to take timely decisions.

2.2 Limitations of Existing System

- Manual analysis takes more time and effort
- Difficult to handle large volumes of hospital data
- Cannot accurately identify disease patterns
- Limited ability to analyze disease distribution based on age and gender
- Difficult to identify key factors influencing disease occurrence
- Lack of automated data analysis techniques
- Limited visualization and reporting capabilities
- Not suitable for real-time healthcare decision-making

2.3 Proposed System

The proposed system uses data mining techniques to analyze hospital data automatically:

- Data preprocessing to clean the dataset, handle missing values, and remove inconsistencies
- Feature selection to identify important attributes such as age, gender, and disease type
- Use of data mining techniques (Decision Tree and Naïve Bayes for classification, K-Means for clustering)
- Identification of frequently occurring diseases
- Analysis of disease distribution across different patient groups
- Visualization of results using charts and graphs
- Improved accuracy and efficiency compared to existing systems
- Scalable and capable of handling large hospital datasets
- Useful for real-world healthcare applications

2.4 Software Requirements

- Python programming language for implementation
- Jupyter Notebook / Google Colab for running and testing the project
- Libraries: Pandas, NumPy, Matplotlib
- WEKA tool for applying data mining algorithms
- Operating System: Windows 10 or above

2.5 Hardware Requirements

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB
- Hard Disk: Minimum 10 GB free space
- Internet connection (for tools and datasets)
- Monitor and keyboard

3. TECHNOLOGIES USED

3.1 Python and Google Colab

- **Python:**
  Python is the main programming language used in this project. It is simple, easy to understand, and widely used for data analysis and data mining. Python provides powerful libraries such as pandas, numpy, and scikit-learn, which are used for data preprocessing, analysis, and implementation of data mining techniques. It also supports data visualization, which is useful for representing disease patterns.

- **GoogleColab:**
  Google Colab is an online platform used to write and execute Python code. It does not require installation and allows users to run code directly in a browser. It provides free computing resources and supports easy dataset uploading and sharing. In this project, Google Colab is used to load hospital datasets, perform preprocessing, apply data mining techniques, and visualize results.

Outcome:
Python and Google Colab together provide a flexible and efficient environment for implementing disease analysis using hospital data.

3.2 Data Mining Techniques

To analyze hospital data and identify disease patterns, the following techniques are applied:

- **DecisionTree:**
  Used for classification of disease data and identifying patterns based on patient attributes

- **NaïveBayes:**
  Used for classification and probabilistic analysis of disease occurrence

- **K-MeansClustering:**
  Used to group patients based on similar characteristics such as age, gender, and disease type

These techniques help in effective analysis, classification, and clustering of hospital data to extract meaningful insights.

3.3 Libraries & Tools

Several Python libraries and tools are used to implement the project:

- **pandas:** For data manipulation and preprocessing
- **numpy:** For numerical computations
- **scikit-learn:** For implementing data mining algorithms and evaluation
- **matplotlib:** For data visualization (charts and graphs)
- **WEKA Tool:** For applying data mining algorithms like Decision Tree, Naïve Bayes, and K-Means
- **Google Colab / Jupyter Notebook:** For writing, executing, and testing Python code

4: SYSTEM ARCHITECTURE

The system architecture of this project represents the overall workflow for analyzing hospital data using data mining techniques. It consists of multiple stages that transform raw patient data into meaningful insights about disease patterns through a structured process.

The first stage is **Data Collection**, where hospital datasets are gathered from reliable sources. The dataset includes attributes such as Patient ID, Age, Gender, and Disease details.

The next stage is **Data Integration**, where different hospital data sources are combined and organized. The datasets are loaded from CSV files using Python tools and structured properly for further analysis.

The third stage is **Data Preprocessing**, which is performed using Python libraries and tools. In this stage, missing values are handled, duplicate or inconsistent data is removed, and categorical attributes such as gender and disease type are properly formatted. Data cleaning and transformation techniques are applied to improve the quality of the dataset.

After preprocessing, the system moves to the **Data Visualization and Analysis** stage. Basic analysis is performed to understand disease distribution, identify patterns, and detect trends. Visualization tools such as charts and graphs help in better understanding of disease patterns.

The next stage is **Data Mining**, where the processed dataset is analyzed using techniques such as Decision Tree and Naïve Bayes for classification, and K-Means clustering for grouping patients with similar characteristics. These techniques help in identifying frequently occurring diseases and patterns.

Following this, the **Model Evaluation** stage is carried out. The analysis results are evaluated using accuracy and other performance measures to ensure reliability and correctness.

Finally, the system produces **Results and Analysis**, where disease patterns, frequently occurring diseases, and patient group distributions are presented. The results are displayed in a structured format, helping healthcare institutions make better decisions.

System Flow Diagram

[Data Collection]

↓

[Data Integration]

↓

[Data Preprocessing]

↓

[Data Visualization]

↓

[Data Mining (WEKA / Techniques)]

↓

[Model Evaluation]

↓

[Results & Analysis]

5: DATA PREPROCESSING

5.1 Sample Data / Records

Age_Group	Gender	BP_Level	Sugar_Level	Cholesterol_Level	Disease	Admission_Type
Young	Male	NormalBP	LowSugar	NormalChol	Malaria	Regular
Old	Female	LowBP	LowSugar	NormalChol	Diabetes	Regular
Middle	Male	HighBP	NormalSugar	NormalChol	Hypertension	Emergency
Middle	Male	LowBP	LowSugar	HighChol	HeartDisease	Emergency
Middle	Male	NormalBP	HighSugar	LowChol	HeartDisease	Emergency
Middle	Female	HighBP	LowSugar	NormalChol	Hypertension	Emergency
Middle	Female	LowBP	NormalSugar	HighChol	Diabetes	Emergency
Old	Female	LowBP	HighSugar	NormalChol	HeartDisease	Regular
Old	Male	NormalBP	NormalSugar	HighChol	Malaria	Regular
Old	Male	NormalBP	LowSugar	HighChol	Malaria	Regular
Old	Male	NormalBP	HighSugar	HighChol	Fever	Emergency
Middle	Male	LowBP	HighSugar	NormalChol	Fever	Regular
Old	Male	NormalBP	LowSugar	NormalChol	HeartDisease	Regular
Old	Male	LowBP	HighSugar	LowChol	Diabetes	Emergency
Old	Male	NormalBP	HighSugar	NormalChol	HeartDisease	Regular
Middle	Female	HighBP	HighSugar	HighChol	Diabetes	Regular
Old	Male	HighBP	HighSugar	LowChol	Malaria	Emergency
Middle	Female	HighBP	HighSugar	LowChol	HeartDisease	Regular
Middle	Female	NormalBP	NormalSugar	LowChol	HeartDisease	Regular
Middle	Male	LowBP	HighSugar	LowChol	HeartDisease	Emergency
Young	Female	LowBP	HighSugar	LowChol	Diabetes	Emergency
Middle	Male	NormalBP	LowSugar	NormalChol	Diabetes	Emergency
Middle	Female	LowBP	LowSugar	LowChol	Fever	Emergency
Old	Female	HighBP	HighSugar	NormalChol	Malaria	Regular
Middle	Female	LowBP	LowSugar	NormalChol	Malaria	Regular
Middle	Male	NormalBP	LowSugar	NormalChol	HeartDisease	Emergency

Figure 5.1 : Sample Data

Note:

- The **Disease column** shows the type of disease diagnosed for each patient
- Different disease values represent various health conditions recorded in the dataset
- The **Age and Gender columns** help in analyzing disease distribution among different patient groups

---

5.2 Dataset Preprocessing

Before applying data mining techniques, the dataset undergoes preprocessing to improve analysis accuracy:

- **Handling Missing Values:**
  Missing data in attributes such as Age or Disease are handled by filling appropriate values or removing incomplete records
- **Data Cleaning:**
  Duplicate records, inconsistent entries, and noisy data are removed to ensure data quality
- **Normalization / Scaling:**
  Numerical features such as Age are scaled to maintain consistency in analysis
- **Encoding Categorical Data:**
  Categorical attributes such as Gender and Disease are converted into suitable formats for analysis
- **Data Transformation:**
  Data is structured and formatted properly for applying data mining techniques
- **Feature Selection:**
  Important features such as Age, Gender, and Disease are selected to improve the effectiveness of analysis

6.SYSTEM MODULES

The proposed disease analysis system is divided into major functional modules. These modules help in data preprocessing, analysis, and result visualization.

6.1 Data Preprocessing Module

- **Purpose:** Prepare the hospital dataset for analysis
- **Functions:**
    - Handle missing values (remove or fill incomplete records)
    - Clean duplicate, inconsistent, and noisy data
    - Convert categorical data (Gender, Disease) into suitable format
    - Normalize/scale numeric values like Age
- **Outcome:** Clean and structured dataset ready for disease analysis

6.2 Feature Selection Module

- **Purpose:** Identify important attributes affecting disease patterns
- **Key Features Used:**
    - Age, Gender, Disease Type
- **Technique:** Selection of relevant attributes to improve analysis accuracy
- **Outcome:** Optimized dataset for effective disease pattern analysis

6.3 Analysis Module

- **Purpose:** Apply data mining techniques to analyze hospital data

- **Techniques Used:**
    - Decision Tree – For classification of disease patterns
    - Naïve Bayes – For probabilistic analysis of disease occurrence
    - K-Means – For clustering patients based on similar characteristics
- **Process:**
    - Load dataset
    - Apply data mining techniques
    - Analyze patterns
- **Outcome:** Identified disease patterns and patient groupings

6.4 Result Generation Module

- **Purpose:** Generate analysis results
- **Process:**
    - Identify frequently occurring diseases
    - Analyze distribution based on age and gender
- **Outcome:** Meaningful insights from hospital data

6.5 Result Display Module

- **Purpose:** Present results in an understandable format
- **Features:**
    - Display disease frequency
    - Show charts (bar graph, pie chart)
    - Present analysis reports
- **Outcome:** Clear visualization for better understanding

7. SYSTEM DESIGN

7.1 Introduction

The system design of the project *Disease Analysis Using Hospital Data* describes the overall structure and workflow of the system. It explains how input data is processed, how data mining techniques are applied, and how results are generated. The design ensures that the system is efficient and capable of handling large hospital datasets.

7.2 Process Flow of the Project

The project follows a step-by-step process to convert raw hospital data into meaningful insights:

- **Data Collection:** Hospital dataset is collected from reliable sources
- **Data Loading:** Dataset is loaded using Python or WEKA
- **Data Preprocessing:** Missing values are handled and data is cleaned
- **Data Transformation:** Categorical data is formatted properly
- **Feature Selection:** Important features like age, gender, and disease are selected
- **Data Mining:** Techniques like Decision Tree, Naïve Bayes, and K-Means are applied
- **Analysis:** Disease patterns and distributions are identified
- **Evaluation:** Results are verified for accuracy
- **Result Generation:** Final insights are displayed

7.3 Flowchart Diagram

The flowchart represents the working of the system in a visual form.

[Start]

↓

[Load Hospital Dataset]

↓

[Data Preprocessing]

↓

[Data Transformation]

↓

[Feature Selection]

↓

[Apply Data Mining Techniques]

↓

[Disease Analysis]

↓

[Clustering (K-Means)]

↓

[Result Generation]

↓

[Display Results]

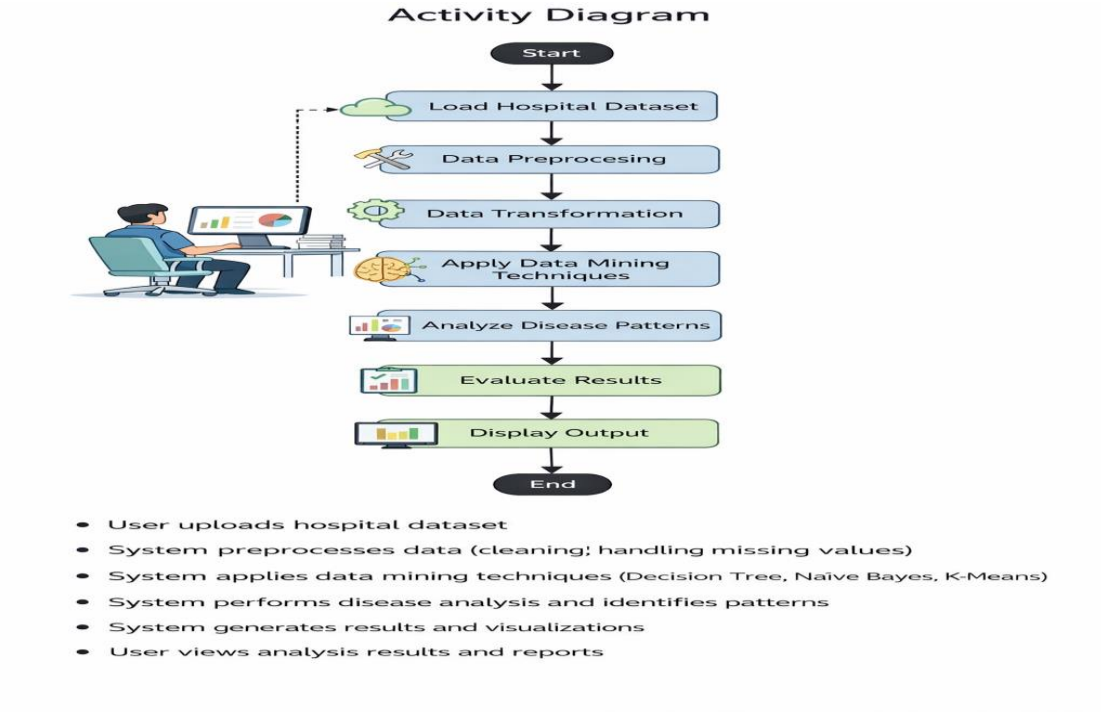
↓

[End]

7.4 UML Diagrams:

Activity Diagram (Description):

- Start → Load Hospital Dataset → Data Preprocessing → Data Transformation → Apply Data Mining Techniques → Analyze Disease Patterns → Evaluate Results → Display Output → End



7.4.Activity Diagram

Use Case:

- User uploads hospital dataset
- System preprocesses the data (cleaning, handling missing values)
- System applies data mining techniques (Decision Tree, Naïve Bayes, K-Means)
- System performs disease analysis and identifies patterns
- System generates results and visualizations
- User views analysis results and reports

8.IMPLEMENTATION

The implementation of disease analysis using hospital data is carried out using Python, Google Colab, and WEKA. The system loads the dataset, preprocesses the data, applies data mining techniques, and displays the results.

8.1 Data Connection / Database

- **Purpose:** Load hospital dataset
- **Implementation:**
  - Dataset stored in CSV file
  - Loaded using Python (pandas) / WEKA tool
- **Outcome:** Dataset is successfully loaded and ready for preprocessing

8.2 Input Validation

- **Purpose:** Ensure that the hospital data is valid and accurate
- **Implementation:**
  - Validation checks on numerical fields such as Age
  - Verification of categorical fields like Gender and Disease
  - Handling missing or incorrect values
- **Outcome:** Clean and valid data improves analysis accuracy

8.3 Exception Handling

- **Purpose:** Handle errors and ensure smooth system execution
- **Implementation:**
  - Manage missing dataset errors and incorrect file paths
  - Handle invalid input values and preprocessing errors
  - Use try-except blocks in Python
- **Outcome:** System becomes stable and reliable

8.4 Testing with Hospital Data

- **Purpose:** Analyze disease patterns using processed dataset
- **Implementation:**
  - Use preprocessed hospital dataset
  - Apply Decision Tree for classification
  - Apply Naïve Bayes for probabilistic analysis
  - Apply K-Means for grouping patients
- **Outcome:**
  - Identification of common diseases
  - Grouping of patients based on characteristics
  - Better understanding of disease distribution

8.5 Training and Evaluating Model

- **Purpose:** Apply and evaluate data mining techniques
- **Implementation:**
  - Apply Decision Tree, Naïve Bayes, and K-Means algorithms
  - Analyze results and verify accuracy
- **Outcome:**
  - Effective disease analysis results
  - Improved reliability of insights
  - Helps in better healthcare decision-making

9. TESTING

9.1 Introduction

Testing is an important phase to ensure that the system works correctly and efficiently. It verifies whether the data mining techniques provide accurate analysis and whether all modules function properly. In this project, testing is performed using hospital datasets to validate disease analysis results and system performance.

9.2 Test Cases

Test Case	Input	Expected Output	Result
1	Valid dataset	Data loaded successfully	Pass
2	Missing values	Cleaned dataset after preprocessing	Pass
3	Patient data	Disease pattern analysis	Pass
4	Dataset analysis	Identification of common diseases	Pass
5	Model evaluation	Accuracy and analysis results	Pass

10. OUTPUT SCREENS

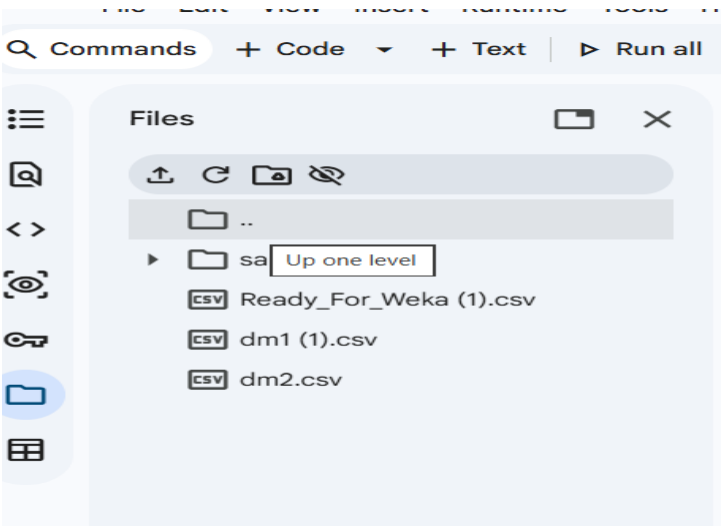
The Disease Analysis System provides output screens showing dataset loading, analysis results, and evaluation details. These screens help in understanding how data mining techniques work on hospital data.

10.1 Data Upload Screen

- **Purpose:** Load hospital dataset for analysis
- **Functionality:**
  - User uploads dataset file (hospital_data.csv)
  - Dataset is read using Python (pandas) or WEKA
  - Displays dataset structure and sample records (Patient ID, Age, Gender, Disease)

Figure 10.1: Hospital Dataset Loaded for Analysis

- **Outcome:** Dataset is successfully loaded and ready for preprocessing



```python
import pandas as pd

# 1. Datasets ni load cheyadam
df = pd.read_csv('dm1 (1).csv') # Review data
data1 = pd.read_csv('dm2.csv') # Customer data

# 2. Duplicates ni handle cheyadam (Data Integration rule)
data1 = data1.drop_duplicates(subset='reviewID')

# 3. Dropna - Deeni valla empty rows unte pothayi
# Kaani be careful: data ekkuva loss avthundante subset=['reviewID'] pettandi
data1 = data1.dropna(subset=['reviewID'])

# 4. Merging (Joining on reviewID)
# 'inner' join vaadithe renditlo common ga unna reviewIDs matrame vastayi
merged_data = pd.merge(df, data1, on='reviewID', how='inner')

# 5. Result ni save cheyadam
merged_data.to_csv("merged_dataset.csv", index=False)

# 6. Result ni display cheyadam
print("Merged Dataset Shape:", merged_data.shape)
```

Figure 10.1: Customer dataset and sentiment dataset loaded for analysis

10.2 Prediction Screen

- **Purpose:** Show analysis results of disease patterns
- **Functionality:**
  - Models applied using Decision Tree for disease classification
  - Naïve Bayes used for probabilistic analysis of disease occurrence
  - K-Means clustering used to group patients based on similar characteristics
  - Analysis performed on test dataset
  - Displays identified diseases and their distribution
  - Shows grouped patient data based on age and gender

• **Outcome:** System predicts customer behavior and classifies reviews automatically

10.3 Reports Screen

- Purpose: Provide structured report of disease analysis performance
- Functionality:

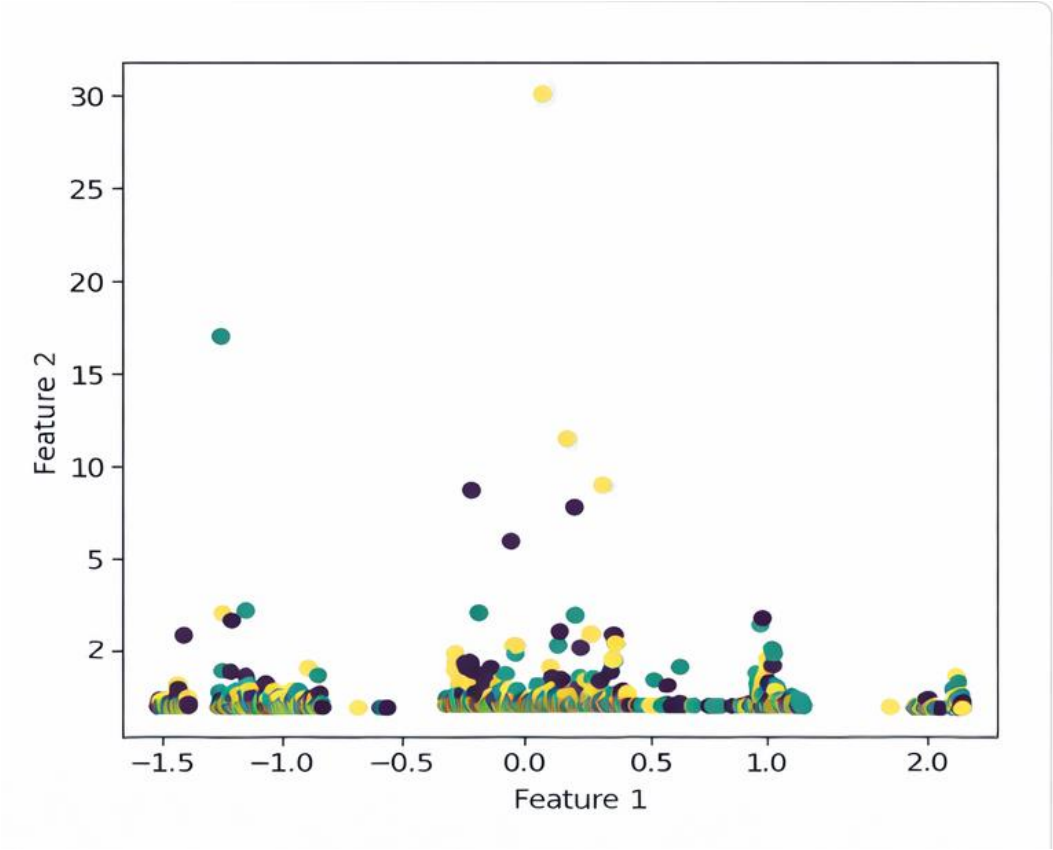
    - Displays analysis accuracy
    - Shows confusion matrix for disease classification
    - Displays classification report including precision, recall, and F1-score
    - Helps evaluate performance of Decision Tree and Naïve Bayes techniques
- Outcome: Clear understanding of analysis performance and result quality

11: Clustering Techniques

11.1 : K-Means Clustering

- Used to divide patients into different groups based on similar characteristics

- Applied using Scikit-learn library
- Data is scaled before applying the algorithm
- K value (number of clusters) is selected
- Patients are grouped into clusters such as similar disease patterns, age groups, or health conditions



11.1: K-Means clustering of customer data

Outcome: Helps in identifying customer segments and possible churn groups

11.2 Hierarchical Clustering

Hierarchical Clustering

- Used to analyze relationships between patients based on similar characteristics such as age, gender, and disease
- Implemented using a dendrogram (tree-like structure) to visualize clustering
- Does not require fixing the number of clusters initially

- Helps in understanding similarities between patient data and identifying disease patterns

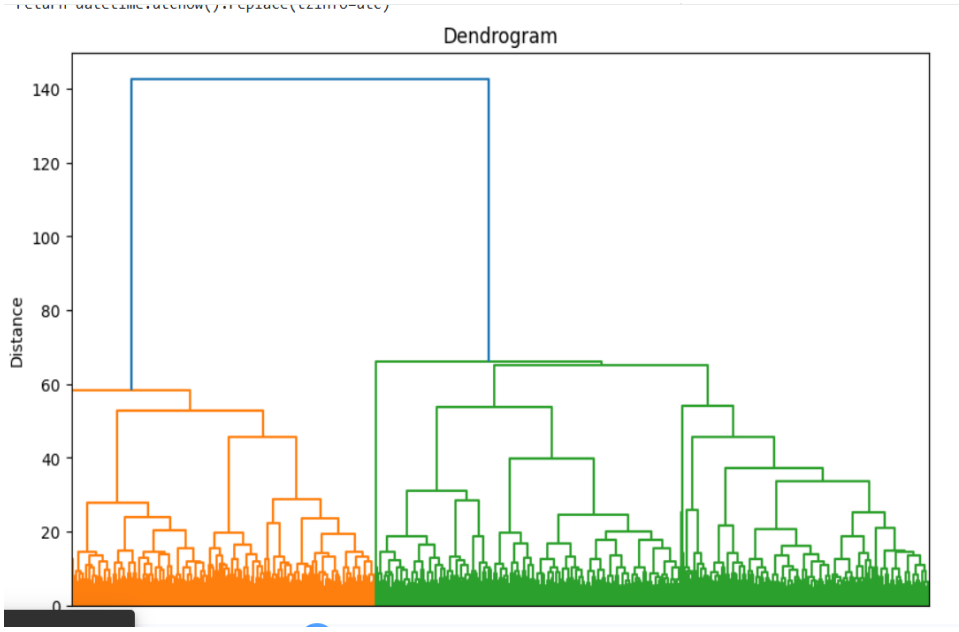


Figure 11.2: Dendrogram representing hierarchical clustering of customer data

Outcome: Provides clear grouping structure of customers

12: CONCLUSION

The *Disease Analysis Using Hospital Data* project successfully developed a data mining-based system to analyze hospital data and identify disease patterns.

Key Points:

- The system accepts hospital datasets and performs preprocessing such as data cleaning, handling missing values, and validation
- Important features such as Age, Gender, and Disease are used for analysis
- Decision Tree algorithm is applied to classify and analyze disease patterns
- Naïve Bayes algorithm is used for probabilistic analysis of disease occurrence
- Clustering using K-Means is performed to group patients based on similar characteristics
- Analysis evaluation ensures accuracy and reliability of results

Outcome:

The project demonstrates how data mining techniques can help healthcare institutions analyze patient data, identify frequently occurring diseases, and understand their distribution. These insights support better healthcare planning, improve resource management, and enhance decision-making in medical systems.

13. FUTURE SCOPE

Several improvements can be made to extend the system:

1. **Interactive GUI:**
   Develop a user-friendly interface for easy dataset upload and visualization of disease analysis results
2. **Web-Based Application:**
   Deploy the system online for real-time access to hospital data analysis
3. **Handling Large Datasets:**
   Enhance the system to process large-scale hospital data efficiently
4. **Advanced Algorithms:**
   Apply advanced techniques such as Support Vector Machine, Neural Networks, and Deep Learning to improve analysis accuracy
5. **Automated Reporting:**
   Generate automatic reports, charts, and dashboards for better understanding of disease patterns
6. **Real-Time Integration:**
   Integrate the system with hospital management systems for live data analysis
7. **Predictive Analysis:**
   Extend the system to predict future disease trends and support preventive healthcare

14. REFERENCES

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